Frame-Based Scene Understanding:

Structured Representations for Introspective Perception in Autonomous Driving

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UC SANTA CRUZ

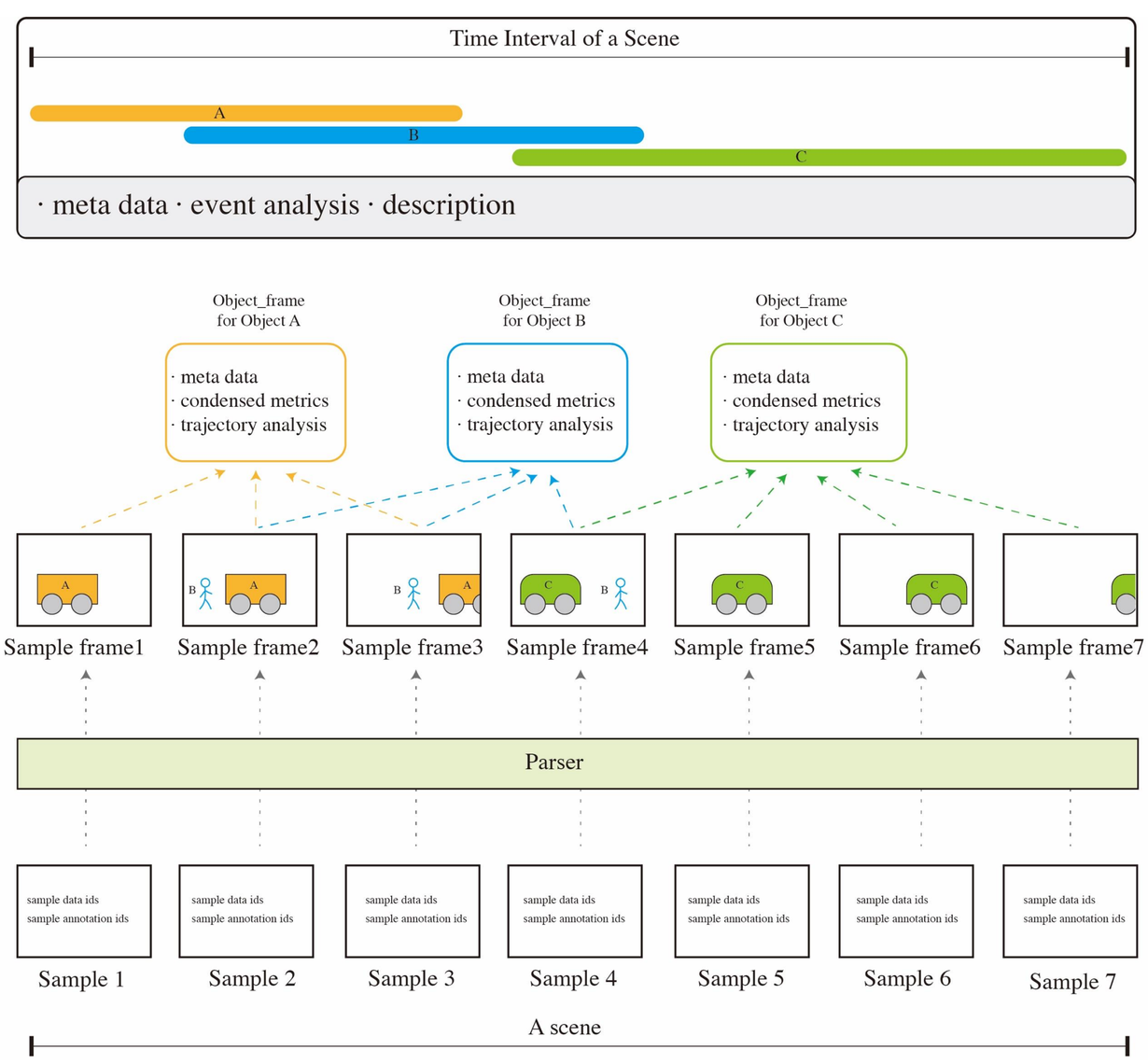


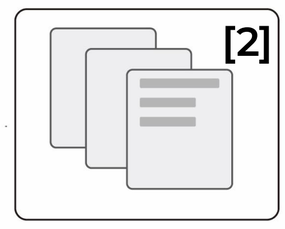
Scene Frame

Object Frame

Sample Frame

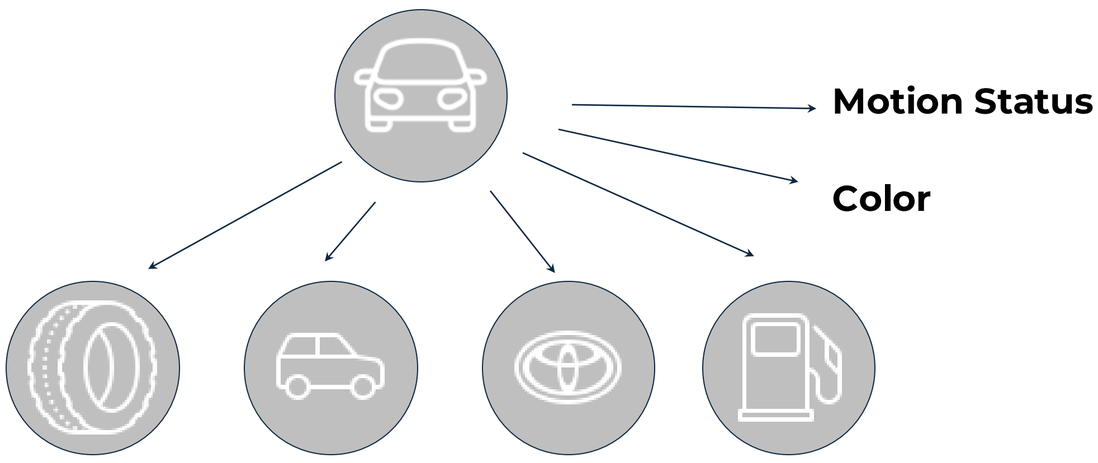
samples (raw data)



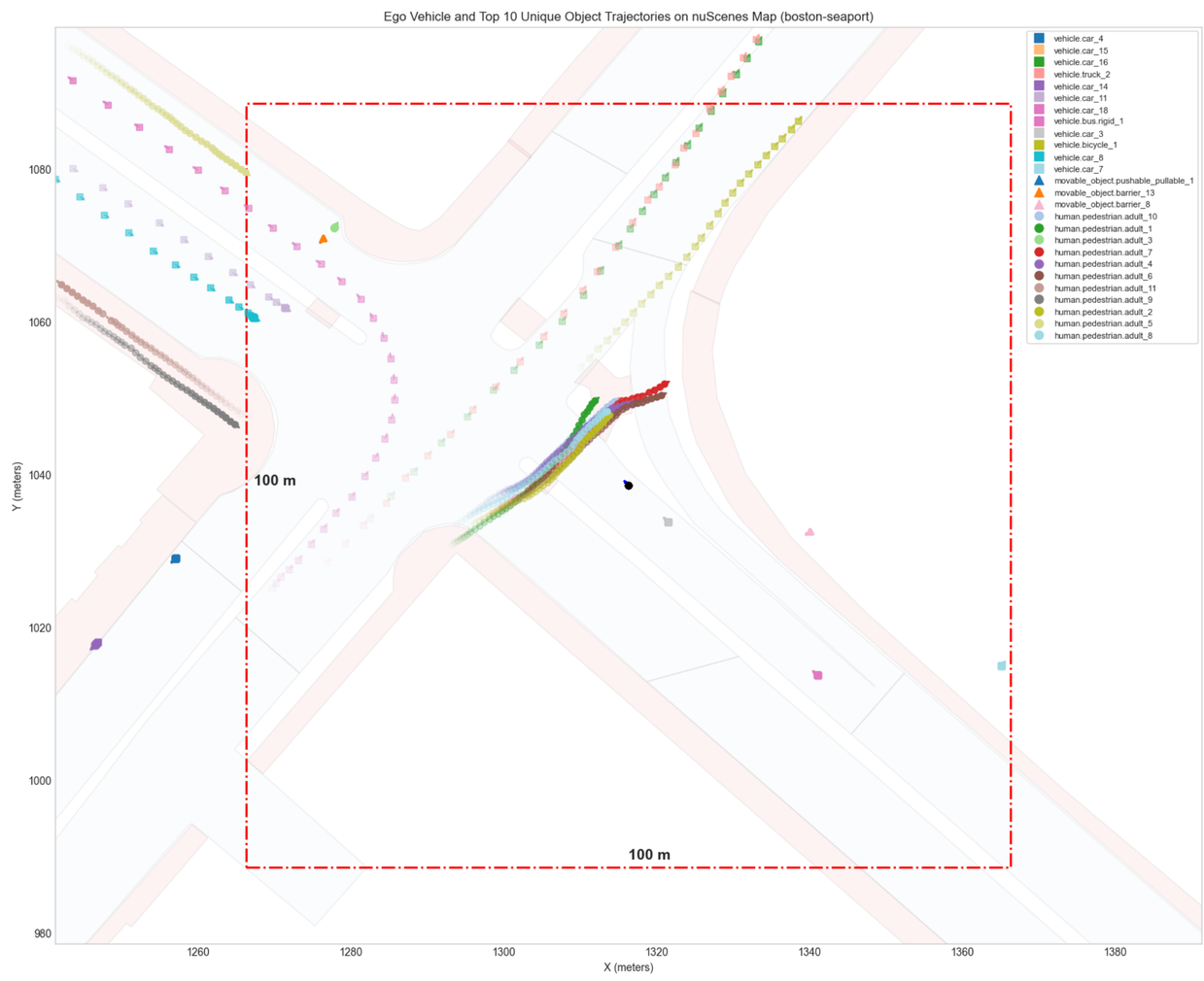


Human scene understanding relies on structured reasoning: we form **expectations** about typical events, focus attention when those expectations are violated, and integrate momentary perceptions into coherent episodes. Our framework captures these core processes through hierarchical Sample–Object–Scene frames, transforming raw sensory data into interpretable, human-aligned representations that support introspective and adaptive perception in autonomous systems.

Frame Theory[1]:



Structure	Type	Make	Power Source
number of doors, wheels	sedan, SUV, truck	Toyota, Ford	gasoline, electric, hybrid



[1] Minsky, Marvin. "A framework for representing knowledge." (1974).
[2] Caesar, Holger, et al. "nuscenes: A multimodal dataset for autonomous driving." CVPR. (2020).